Messaging Implementation Guidelines

Global (excluding The Americas)

Version 1.3

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Preface

This document provides technical information regarding the implementation of mBlox's mobile terminated (MT) and mobile originated (MO) messaging services. If it does not answer the Client's question please contact mBlox customer support, (see **'Where to get help**' below).

The mBlox platform provides a functionally rich, robust and scalable bi-directional SMS delivery service, through which service providers may access SMS messaging products. Service providers integrate to the platform using one of the available mBlox interfaces.

This guide is applicable for clients using mBlox's WholesaleTxt, GlobalTxt, WholesaleTxt Direct and EnterpriseTxt products and should be used in conjunction with the technical manual for the client's chosen interface type. It includes details on how the service operates and the specific use of certain interface parameters.

mBlox's messaging products provide a full service solution for launching standard rate SMS applications, including:

- Enabled Two-way, Long Number based SMS messaging with subscribers of Global mobile networks, excluding The Americas.
- Enabled Two-way, Short Code based SMS messaging with subscribers of Global mobile networks, excluding The Americas.

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Where to get help

For any further information or support please visit our website at **www.mblox.com** or contact one of our offices:

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Section 1: Change History

1.1 Current document

Document Title: mBlox Messaging Implementation Guidelines – Global (excluding The Americas)
Author: Malin Jason
Classification: Confidential
Version: 1.3
Date of Issue: 12th of June 2013

1.2 Version History

VERSION No.	DATE	REASON FOR ISSUE
1.0	15th Sep 2012	Release
1.1	22nd Oct 2012	Product Addition
1.2	29th Nov 2012	Long Number expansion
1.3	12th of June 2013	Product Addition, Long Number and Short Code expansion , removal of the MNC/MCC section

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Section 2: Products Overview

2.1 WholesaleTxt

WholesaleTxt is a messaging service designed for mobile aggregator experts, providing SMS connectivity to mobile subscribers via network operators ("Network Operators") globally[1]. An optional reply path is supported via Long Numbers which may be leased by the Customer on a country-by-country basis. The countries and Network Operators that are supported by this product, along with the restrictions which apply to them are defined in the mBlox Messaging Products Reach List.
This Product uses a combination of high quality providers, direct Network Operator connections and mBlox's SMSC to provide advanced message features and least cost routing for Customers' messages.
Clients are provisioned with one mBlox account with a single profile ID. This allows outbound and inbound traffic.

2.2 GlobalTxt

GlobalTxt is a messaging service providing SMS connectivity to mobile subscribers via network operators ("Network Operator") globally[1]. An optional reply path is supported via Long Numbers which may be leased by the Customer on a country-by-country basis. The countries and Network Operators that are supported by this product, along with the restrictions which apply to them are defined in the mBlox Messaging Products Reach List.
This Product uses a combination of high quality providers, direct Network Operator connections and mBlox's SMSC to provide advanced message features and best value routing for Customers' messages.
Clients are provisioned with one mBlox account with a single profile ID. This allows outbound and inbound traffic.

2.3 WholesaleTxt Direct

WholesaleTxt Direct is a messaging service for mobile aggregator experts, providing SMS connectivity to mobile subscribers via Network Operators in countries in EMEA and APAC. An optional reply path is supported via Short Codes and Long Numbers which may be leased by Customer. The general features of this product are defined in Section 4, but restrictions varying by Network Operator may apply. The countries and Network Operators that are supported by this product, along with the restrictions which apply to them are defined in the mBlox Messaging Products Reach List.
This Service uses a combination of direct Network Operator connections, one-hop supplier connections and mBlox's own SMSC to provide advanced message features and priority routing for Customers' messages.

[1] Excluding The Americas.

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2.4 EnterpriseTxt

EnterpriseTxt is a messaging service designed for enterprises and brands and makes it easy to send and receive SMS from millions of subscribers across multiple countries in APAC and EMEA in a fast, reliable and secure way. EnterpriseTxt is a two-way enabled messaging service providing SMS connectivity to mobile subscribers via Short Codes or Long Numbers. This Service uses a combination of direct Network Operator connections, one-hop supplier connections and mBlox's own SMSC to provide advanced message features and priority routing for Customers' messages.

Timestamp

In addition EnterpriseTxt offers guarantee of message delivery with local subscribers' Time Zone via the EnterpriseTxt Timestamp variant, which is offered per region.

EnterpriseTxt Americas

For information regarding EnterpriseTxt Americas, please refer to the mBlox Messaging Guidelines – Americas or contact your mBlox Sales representative.

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Section 3: SMS Messages

3.1 Message Types

All mBlox products offer full support for one-way (MT) or two-way (MT&MO) SMS messaging.

All mBlox Messaging Products offer full two-way consumer interaction via mBlox Long Numbers solution (detailed in Section 8) and EnterpriseTxt and WholesaleTxt Direct also offer full two-way interaction via network operator standard rate Short Codes (detailed in Section 7).

SMS Messages	Description
Standard Rate Mobile Terminated (MT) Message	Sent from the client's application to a handset. These are free to the end user.
Standard Rate Mobile Originated (MO) Message	Sent from the handset to the client's application at standard rate or as determined by the operator's pricing policy.
Delivery Notification	Sent to the client's application in response to an MT message, providing details on the status of the message. Intermediate status indicates the message is in the operator network and delivery is being attempted. Final status indicates the final state of the message.

Table 1 – Message Types

3.2 Standard Rate Messages

Standard rate messages are billed by the network operator against the end user's telephone charges, or deducted from their message bundle, according to the user's data plan. This includes the sending of MO messages.

3.3 Accounts on the mBlox Platform

The client is provisioned with one or two accounts[1] on the mBlox messaging platform.
Each account supports different message types.

[1] A second account is provided for EnterpriseTxt and WholesaleTxt Direct for MO messages.

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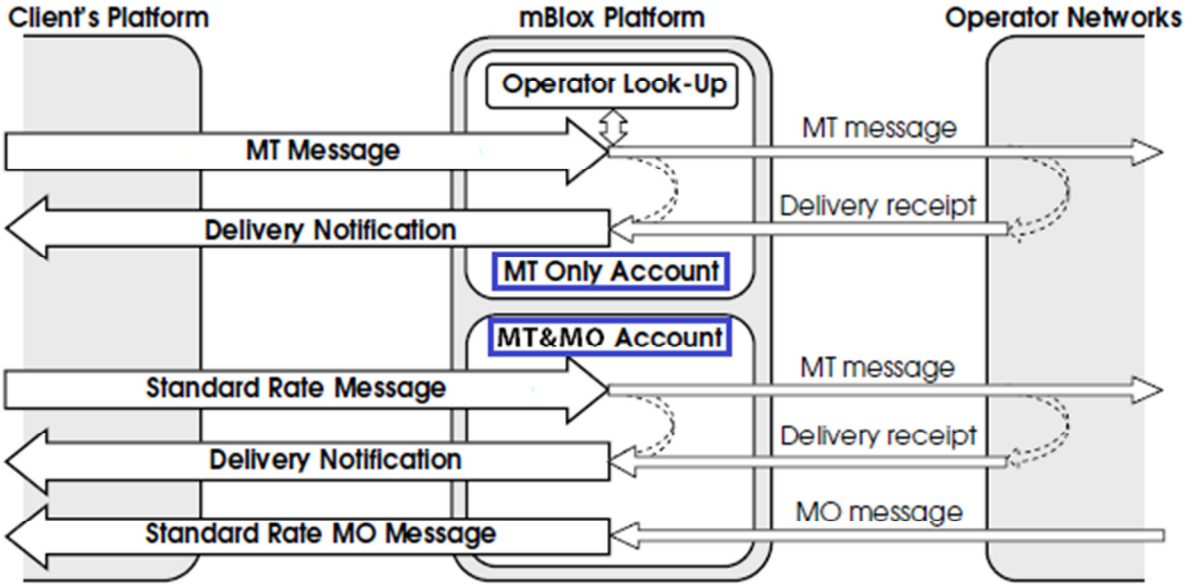

The MT Account supports:

- MT Messages
- Delivery Notifications related to these messages

The MT&MO Account supports:

- MT messages
- Delivery Notifications related to these messages
- Standard Rate MO messages via Short Code

Each account has a separate account name, password and Profile ID, which are provided by mBlox during provisioning.

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Section 4: Product Features

Product Features	WholesaleTxt	WholesaleTxt Direct	GlobalTxt	EnterpriseTxt	EnterpriseTxt TS
Numeric Originator	Yes	Yes	Yes	Yes	Yes
Alpha-Numeric Originator	Yes	Yes	Yes	Yes	Yes
Short Code Originator	Yes	Yes	Yes	Yes	Yes
Unicode	Yes	Yes	Yes	Yes	Yes
Binary	Yes	Yes	Yes	Yes	Yes
Flash SMS	Not guaranteed (1)	Not guaranteed (1)	Not guaranteed (1)	Not guaranteed (1)	Not guaranteed (1)
Wap Push	Yes	Yes	Yes	Yes	Yes
Concatenated Messages	Yes	Yes	Yes	Yes	Yes
Delivery Notifications	Yes (2)	Yes (2)	Yes (2)	Yes (2)	Yes (2)
MNP Compliant	Yes	Yes	Yes	Yes	Yes
Local Character Set	Not guaranteed (1)	Not guaranteed (1)	Not guaranteed (1)	Not guaranteed (1)	Not guaranteed (1)

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Local Timestamp	Not guaranteed	Not guaranteed	Not guaranteed	Not guaranteed	Yes
Validity Period	24 hours	48 hours	48 hours	72 hours	72 hours
Throughput	10 sms/s	20 sms/s	20 sms/s	30 sms/s	30 sms/s
Latency	Not guaranteed	<30 seconds	<30 seconds	<10 seconds	<10 seconds
Product Resilience	Always Available	Always Available	Always Available	Always Available	Always Available
	(Failover available)	(Failover available)	(Failover available)	(Failover available)	(Failover available)
Interfaces: MSIP (through JAVA API), SMPP, XML	Yes	Yes	Yes	Yes	Yes
Quote currencies	AU$/£/€/US$	AU$/£/€/US$	AU$/£/€/US$	AU$/£/€/US$/SEK /SGD	AU$/£/€/US$/SEK /SGD

[1] This feature is not included in the product but may be available dependant on destination. mBlox does **not** guarantee the availability or offer support for the feature

[2] mBlox only works with operators that support Delivery Notifications (DNs). As mBlox does not receive DNs back from the operators in 100% of cases, mBlox cannot guarantee that the delivery status for all messages will be returned to the client. mBlox will pass on all delivery status information received from operators.

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4.1 Numeric / alphanumeric / Short Code originators

The originator on an MT is displayed as the sender field (**From:**) on the handset. This field can be used to allow subscribers to reply to the outbound SMS to a Short Code, Long Number or to indicate the service which generated the message. If an originator is not supplied when sending a message a default value will be provided and appear as the sender of the message received by the subscriber.

a. Numeric originator is the default originator type of an outbound SMS and uses the standard format for international MSISDNs, example '447812121212'. The maximum length of a numeric originator is 15 chars. This originator type is reply-able from the handset and typically displays including a preceding plus sign, as '+447812121212'.

b. Short Code originator is also a numeric value but is displayed as a network Short Code without preceding plus sign, which allows the use of this service in conjunction with national SMS Short Codes. The maximum length of a Short Code originator is 15 chars.

c. Alphanumeric originator can contain characters from a-z, A-Z, 0-9 and '_' (underscore). Other characters may cause the message to fail. Subscribers are not able to reply to alphanumeric originators.

The client should refer to the relevant interface technical manual for the required parameters to set each originator type.

4.2 Flash Messages

A Flash SMS is displayed immediately on the screen upon arrival, without the need for the subscriber to actively view it. Flash SMS are typically not saved in the inbox and will be deleted once viewed unless actively saved by the subscriber. The client should refer to the relevant interface technical manual for the required parameters to set this message type (note that the relevant parameter for the Java API is "**smsClass**").

The Flash SMS feature availability is dependent on support in the sending SMSC and in the receiving phone and, hence, may not always be available.

4.3 Unicode

When characters outside the standard supported character set (Section 3.8) are required, the message has to be sent as Unicode encoded as UTF-16. As Unicode uses 16 bits per character the message length is limited to 70 characters. Unicode is typically used for Arabic, Greek and Asian character sets. The character set has to be supported by the handset to display correctly.

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4.4 Binary

Binary SMS are used to deliver rich content such as EMS, ringtones, wallpapers, OTA WAP settings, phonebook entries, WAP push notifications and more. Other content types can be delivered using the UDH header according to manufacturer specifications. The client should refer to the relevant handset manufacturer's documentation for more information.

4.5 Concatenated Messages

Concatenated messages allow subscribers to receive messages longer than the standard short message (**160 characters for GSM**). This feature allows for longer messages to be send as numbered parts to be concatenated by the receiving handset. Depending on the handset model these will be displayed as several SMS, numbered as parts of a longer message or as a normal SMS.
To submit a concatenated message, the UDH parameter needs to be set according to the handset manufacturer's specifications. The inclusion of a UDH reduces the maximum message body length, as the overall payload length of message body plus UDH has a fixed maximum.

The originator has to be the same for all parts of the message for the message to be reassembled correctly by the handset.

4.6 Delivery Notifications

Delivery notifications are generated on a per message basis, providing details about the delivery status of the message in the operator network. The status that is returned falls into one of two categories:

- **Intermediate status** - The message is in storage within the operator network and delivery is being attempted
- **Final status** - Indicates the final status of the message.

In normal operation, every message which has been accepted by the mBlox platform will generate a final status delivery notification. In addition, one or more intermediate status delivery notifications may be generated. If no delivery notification is received from the operator network, mBlox will return a final status of **Lost notification/Unknown**.

The client should refer to the relevant interface technical manual for details about the various delivery notification statuses and codes.

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4.7 Local character sets

Support for plain text characters is limited to the subset of the **Latin-1** (**ISO-8859-1**) character repertoire which is shown in Table 2. Details on how to encode these characters can be found in the relevant interface technical manual. This character set is sufficient for messaging in most countries using the Latin alphabet.

Local characters outside the table below may fail due to not being supported by the sending SMSC or the receiving phone. A wider character set can be used by sending and encoding the message as Unicode.

										LF			CR		
SP	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/
0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
P	Q	R	S	T	U	V	W	X	Y	Z					_
	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
p	q	r	s	t	u	v	w	x	y	z					
	¡		£	¤	¥		§								
															¿
				Ä	Å	Æ	Ç		É						
	Ñ					Ö		Ø				Ü			ß
à				ä	å	æ		è	é			ì			
	ñ	ò				ö		ø	ù			ü			

Table 2 – Supported plain text characters

4.8 Local Timestamp

For certain applications, such as time reporting systems, fleet management systems or alert systems, it is critical that the message is time stamped using the local time zone to allow for accurate use on the recipient end.

The local timestamp feature is available with mBlox EnterpriseTxt Timestamp.

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The client should contact their account manager for further information. On products not listed as supporting local timestamp, the message timestamp may vary with the local time zone of the originating SMSC used by mBlox to send the message.

4.9 Validity Period

The validity period of a message is the period of time for which the message will be stored and retried by the operator if the destination handset is not immediately available. After this period has expired the message will no longer be retried.
The validity period of each message depends on **two factors**.

- The **first** is the default validity per product that can be found in the product features, Section 4 of this document.
- The **second** is a parameter which may be used to set a shorter validity for messages that have no relevance unless delivered within a short timeframe, e.g. goal-by-goal results for sports events.

The validity period cannot be set to be longer than the product default maximum. The client should refer to the relevant interface technical manual for information on how to specify the validity period.

4.10 Throughput

Throughput defines the maximum sustained number of messages per second which mBlox guarantees to process. The default throughput value offered per product is detailed in Section 4 – Product Features.
Higher throughput may be available for certain destinations on request. The client should contact their account manager if a higher throughput is required.

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Section 5: Parameter Usage

Product	Message Type	Originator	Originator Type	Profile ID	Destination	Operator
EnterpriseTxt WholesaleTxt Direct	Standard Rate MT	Numerical, Short Code or Alphanumerical	1, 3, 5	Provisioned by mBlox	End-user MSISDN	Not present
	Standard Rate MO	End-user MSISDN	Not present	Not present	Standard Rate Short Code	Originating Operator ID
	Delivery Notification	End-user MSISDN	Not present	Not present	MT originator (only if Numeric or Short Code)	Originating Operator ID*
WholesaleTxt, GlobalTxt, EnterpriseTxt	Standard Rate MT	Numerical, Short Code or Alphanumerical	1, 3, 5	Provisioned by mBlox	End-user MSISDN	Not present
	Standard Rate MO	End-user MSISDN	Not present	Not present	Alphanumerical	Originating Operator ID
	Delivery Notification	End-user MSISDN	Not present	Not present	MT originator (only if Numeric or Short Code)	Originating Operator ID*

* Please contact your Account Manager to enable Originating Operator ID in Delivery Notifications.

The following parameters need to be sent in MT messages:

- Destination
  - Will be the consumer's MSISDN
- Originator
  - Will be the 5 digit Short Code, Long Number or Alphanumeric displayed to the consumer receiving the message. The parameter is Customer specific and will be provided when your account is provisioned[1].
- Originator Type

[1]Only applicable to Short Codes and Long Numbers.

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- Should be set to the relevant value, depending on the product used. 3 for Short Codes, 1 for Numeric and 5 for Alphanumeric originators.

- Profile ID
  - The ProfileID will be unique per product and provided when the Customer account is provisioned.

The following parameters are included in MO messages:

- Destination
  - Will be the 5 digit Short Code or Long Number consumers are sending their messages to. The parameter is Customer specific and will be provided when your account is provisioned.
- Originator
  - Will be the consumer's MSISDN that sent the message
- Operator ID
  - The originating subscriber's Operator ID may not always be provided. When included it will be in the form of MCC / MNC (Mobile Country Code / Mobile Network Code).
  - Please refer to the mBlox Reach List for all MCC/MNC codes.

The following parameters are included in Delivery Notifications:

- Operator ID
  - The originating subscriber's Operator ID is provided upon request.
  - The Operator ID is in the form of MCC / MNC (Mobile Country Code / Mobile Network Code).
  - Please refer to the mBlox Reach List for all MCC/MNC codes.
    - If an Operator ID value is returned that is not in mBlox Reach List, The Operator ID is internal and the SMS is sent through an unknown deliverer.

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Section 6: Geographical Exceptions

6.1 France

This section applies to clients sending traffic to France via EnterpriseTxt.

6.2 Universal Stop Command

In accordance with French regulations and guidance (www.smsplus.org), end users must be able to prevent the sending of future messages by sending the word "STOP" to the message sender.
"STOP" text, is any MO text message starting with the word 'stop'. The word stop can be made up of any mix of lower or upper case letters and be followed by a space, full stop or by no further character.
The industry must make reasonable efforts to understand and process a "STOP" command where stopping the service is clearly the user's intention.
To facilitate our client's regulatory compliance mBlox has implemented functionality to support the universal STOP command and ensure that we forward all STOP messages to its MT clients who manage services in France.

6.3 Universal Stop Command – Shared Short Codes

Clients sending direct traffic to France have the ability to order keywords for two way interaction with their end users. End users must be able to send in a 'stop' message to these messages. mBlox has implemented shared Short Code functionality so that each time a message containing the body, (this is not case sensitive) "STOP" is received on a shared Short Code, it is forwarded to all clients using that code.

Upon receipt of a "STOP" from a Shared Short Code

- The client must check its records and establish if the MSISDN is known to their application
- If the MSISDN is known the client application must "STOP" sending messages to that number relating to the last service used.
- If the MSISDN is unknown to the application the STOP command request must be discarded. By no means must the client application send out any MTs to that MSISDN

All confirmation "STOP" messages must be sent free to the user.

A mobile originated "STOP" request must never be interpreted as an opt-in. That is, no communications may be sent to an unknown/ unregistered MSISDN. Opting in MSISDN's in this manner is strictly forbidden.

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"STOP" sent in response to a marketing message must unsubscribe the user from further marketing messages. In addition such user's MSISDN's must not be sold on to other companies for marketing purposes.

6.4 Free France

All messages sent on EnterpriseTxt to carrier Free will be overwritten with a Short Code originator. Messages with dynamic originators can be sent but are subject to restrictions. Marketing Messages must be set up on a separate profile and the message body must contain the following sentence: "STOP SMS au 36111". If not, Free will reject the message. Notification/Alert Services also needs to be set up on a separate profile but the message body is not monitored by Free.

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Section 7: Short Codes

mBlox offer Short Codes as an optional response path on EnterpriseTxt and WholesaleTxt Direct. The solution provides businesses with standard rate Short Codes that acts as a reception mechanism for consumer generated SMS messages (termed Mobile Originated, or MOs). A Short Code is normally 4 or 5 digits long and is easy for consumers to remember. Short codes have national reach, i.e. UK short codes can only be reached via mobile phone numbers starting with +44.
mBlox provides Short Codes in various countries (see below) and offer unique Short Codes, keywords on Shared Short codes or Smart Sessions on shared Short Codes (see below).

The product offers optional full two way consumer interaction via Standard Rate Short Codes in the following countries:

- UK
- Sweden
- Ireland
- Spain
- Belgium
- France[1]
- New Zealand
- Netherlands
- Germany

Clients who wish to take advantage of this offer will be provisioned with one account to send all MT traffic and one to receive MO messages, depending on the chosen country.

When used in conjunction with one of mBlox's leading Mobile Terminated (MT) messaging products (EnterpriseTxt) Short Codes enable full two-way consumer interaction. Whether used for straightforward customer service applications or high volume customer response events, mBlox provides a premier solution for companies seeking reliable customer interactivity.

7.1 France

Services on unique Short Codes in France must be initiated via an MO Message to the Short Code on which the service is hosted. From the MO message, the client will be able to determine the operator on which the end user is subscribed and an alias by which they can be identified, but not their MSISDN phone number. An alias is a 12 to 17 long number used by the carriers to replace the MSISDN. There is

[1] This country requires a separate account for unique short-codes to send and receive messages; please contact your mBlox Sales representative for further details.

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a 1-1 mapping between the alias and a sim card. The alias is operator specific and the life time of an alias may vary depending on carrier.

The following parameters need to be sent in MT messages:

- Destination
  - End User Alias
- Originator
  - 5 digit Short Code. The parameter is Customer specific and will be provided when your account is provisioned[1].
- Originator Type
  - 3 – Short Codes
- Operator
- Tariff
  - "0"
- Profile ID
  - The ProfileID will be unique per product and provided when the Customer account is provisioned.

The following parameters are included in MO messages:

- Destination
  - 5 digit Short Code
- Originator
  - Alias
- Operator

7.2 Smart Session

Sticky Session is designed for clients that wish to use shared short codes to receive SMS without using keywords. When the client submit an MT request, a session opens on the mBlox platform. When the end-user responds to the MT with an MO, the MO is forwarded to the client and the session is closed. The Session is open for 24 hours per default.

Feature Restrictions:

- If two of mBlox clients send two messages to the same end-user from the same short code within the same time period, the MO will be sent to the second client.
- If the same MO message matches both a keyword and an active sticky session, the SMS will be sent to the keyword client
- Sticky Session is only enabled for short code originators

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- mBlox responds to STOP messages on shared Short Codes. Clients that want to respond to these messages themselves (only with Smart Session enabled), may request this from the Account Manager.

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Section 8: Long Numbers

mBlox Long Number solution provides businesses with a virtual mobile telephone number that acts as a reception mechanism for consumer-generated SMS messages (termed Mobile Originated, or MO, messages). Known as Long Numbers, these virtual phone numbers conform to the numbering system of a specific country and function like a standard phone number. They can be reached by users located in either that country or internationally, allowing businesses to develop one service with global participants. Compared to national Short Codes, which have country-specific reach, can be expensive and require more administration Long Numbers can offer businesses a cost-effective solution with increased flexibility. For example, by using a UK-based Long Number starting with the UK country code +44, mBlox clients do not incur the associated Short Code charges or registration processes.

When used in conjunction with one of mBlox's leading Mobile Terminated (MT) messaging products, Long Numbers enable full two-way consumer interaction. Whether used for straightforward customer service applications or high volume customer response events, mBlox provides a premier solution for companies seeking reliable customer interactivity in a cost-effective manner.

8.1 Technical Overview

Long Numbers are available with mBlox WholesaleTxt,WholesaleTxt Direct, GlobalTxt and Enterprise Txt products.
Clients will also be provided with the normal technical information when they are provisioned including:

- Username
- Password
- Long number/s
- Connection information which is interface dependent.

The specific detail of what an MO message will look like is interface protocol dependent. Clients are advised to refer to the specific interface specification for details. Generally however each Long Number MO message will contain:

- Originator – MSISDN of the end user who sent the MO
- Destination – Client's Long Number
- Operator – MCC/MNC code (this information is not always provided)
- Body – body of the message the end user has sent in

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Clients are able to route Long Numbers to different socket connections on their platform by use of different socket names. These need to be provisioned and ordered in advance.

8.2 Long Number Features per Country

This section details the country specific Long Number solutions offered by mBlox. mBlox currently offer Long Number solutions with local numbers in

- Australia
- India
- Ireland
- Italy
- Jersey
- Norway
- Poland
- South Africa
- Sweden
- UK

Most of these Long Number solutions have local reach, with the exception of UK and Jersey. Please see the 'mBlox Reach List' for specific details about the operators reach per solution.

All Long Number countries provide unique Long Numbers with the exception of South Africa where a pool of Long Numbers are offered. Please see below for further information.

This document should be used in conjunction with the specific interface implementation guide that details the format of mobile originated messages received.

8.3 Geographical Exceptions

UK

MO messages sent from Australian handsets to UK long numbers should have the destination (long number) in one of the following formats: either +447797800180 or 0011447797800180. When the leading 00 are submitted such as 00447797800180, MO messages may not get delivered.

South Africa

The South Africa Long Number solution contains a pool of Long Numbers. The mBlox client will receive a unique Long Number originator that is submitted to mBlox as originator. The originator will

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be semi randomly overwritten and sent to the end-user. When the end-user responds to the message, the original originator will be switched back and forwarded to mBlox Client.

8.4 Long Numbers - Local character sets

Support for plain text characters is restricted to the subset of the **Latin-1** (**ISO-8859-1**) character repertoire which is shown in **Table 3**. Details on how these will be encoded can be found in the relevant interface technical manual.

Please note that characters may vary slightly depending on the Long Number provider.

										LF			CR		
SP	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/
0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
P	Q	R	S	T	U	V	W	X	Y	Z					_
	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
p	q	r	s	t	u	v	w	x	y	z					
	¡		£	¤	¥		§								
															¿
				Ä	Å	Æ	Ç		É						
	Ñ					Ö		Ø				Ü			ß
à				ä	å	æ		è	é			ì			

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	ñ	ò				ö		ø	ù			ü			

Table 3 – Supported plain text characters

8.5 Feature Table

The table below summarises the features available on the various Long Number solutions that we offer. Where concatenated message support is listed please note that these messages will come through as separate mobile originated messages with the appropriate UDH (User Data Header) for the message.

Product	Australia	India	Ireland	Italy	Jersey	Norway	Poland	Sweden	UK	South Africa
Country Code	+61	+91	+353	+39	+44[1]	+47	+48	+46	+44	+27
International Reach	Yes	No	No	No	Yes	No	No	No	Yes	No
Vanity Numbers	Yes	No	No	No	No	No	No	No	No	No
Message Format	Standard and concatenated text, binary and Unicode	Standard text message	Standard and concatenated text, binary and Unicode	Standard text message	Standard and concatenated text, binary and Unicode	Standard and concatenated text, binary and Unicode	Standard and concatenated text, binary and Unicode	Standard text message	Standard text message	Standard text message
Throughput	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second	Up to 10 SMS per second
Interfaces: MSIP (through JAVA API),	MSIP	MSIP	MSIP	MSIP	MSIP	MSIP	MSIP	MSIP	MSIP	MSIP

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SMPP, XML										

[1]The cost of sending an SMS to a Jersey-based phone number for a mainland UK end-user is set by the user's network operator and may be classified as an international message by certain networks, even though the country code is +44.

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